

TrES-3b

R-0452

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220622 · created 2026-07-19T17:35:24+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459752.8742 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.027
Transit depth (Rp/Rs)^2	3.71 +/- 1.02 [%]
Orbital Inclination (inc)	81.93 +/- 1.2
Airmass coefficient 1 (a1)	1.093 +/- 0.013
Airmass coefficient 2 (a2)	-0.07 +/- 0.026
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#2 - [623, 369]
Optimal Aperture	3.63
Optimal Annulus	18.01
Transit Duration (day)	0.057 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.027 vs 0.16309 (deviation 1.11σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.03σ)
Mid-time O–C	-5.0 min
Depth SNR	7.1
Residual scatter	1.19 %

Light curve

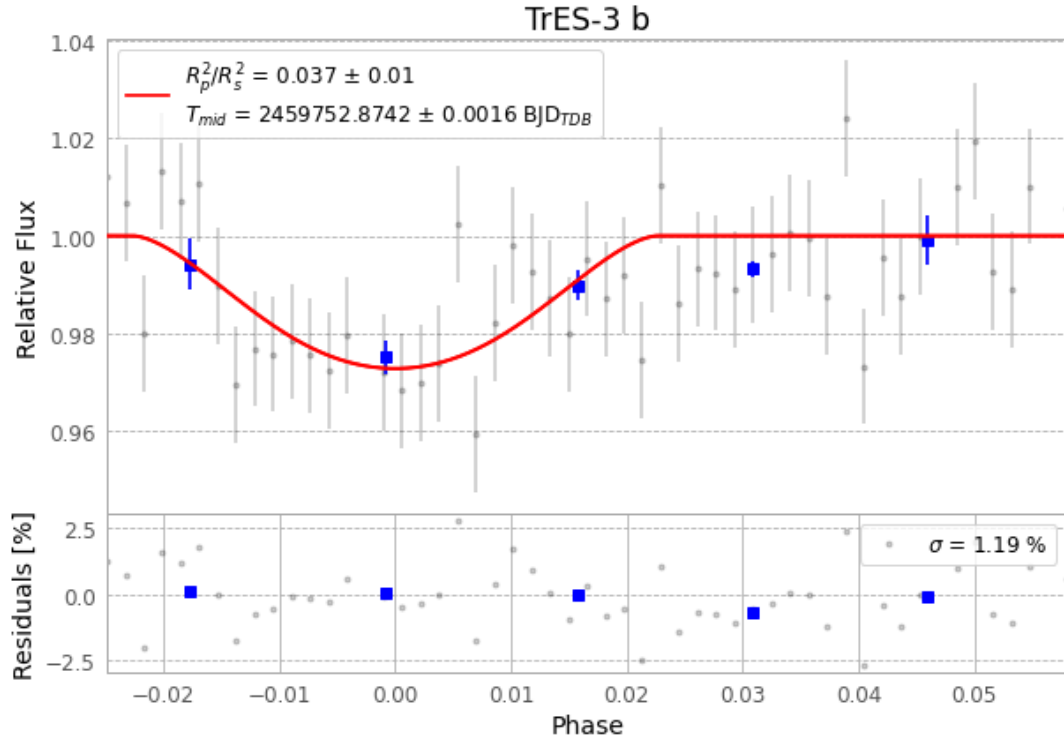


Figure 1 — FinalLightCurve_TrES-3 b_2022-06-22.png (SHA-256 9dd79e226094011f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [323, 335], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6db751123bc432bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

51 FITS frames (33 MB), TRES-3220622080615.FITS → TRES-3220622104216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220622.log (e13e3efa9930...), FinalLightCurve_TrES-3 b_2022-06-22.png (9dd79e226094...), FinalLightCurve_TrES-3 b_2022-06-22.csv (649f7b302e4c...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 17:35Z.